Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Chemistry Examination May 2017

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC-II) Chemistry. It contains comments on candidates' responses to the 2017 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Comments

Overall candidates showed good progress in this exam. In Constructed Response Questions (CRQs), they displayed better understanding of the concepts such as electronic configuration, nucleophilic substitution reactions, ozonolysis, halogenation of alkene, preparation of phenol, and functional group chemistry. Similarly, they represented good knowledge of the concepts related to environmental and analytical chemistry i.e. greenhouse effect and mass spectrometry. On the other hand, they showed little interest and need more focus in compounds of s-block elements, chemistry of transition elements, IUPAC and chemistry of carbonyl compounds.

In Extended Response Questions (ERQs), candidates showed average performance. They found difficulty to represent the stepwise mechanism for the formation of amide and the reactions of carboxylic acid. They performed slightly better to describe the addition reactions of ethen, but still struggled to mention the conditions or stepwise addition where required. Similarly, they need to focus more on concepts related to biochemistry and industrial chemistry as they showed limited understanding towards protein, dyes and pesticides.

Detailed Comments:

Constructed Response Questions (CRQs)

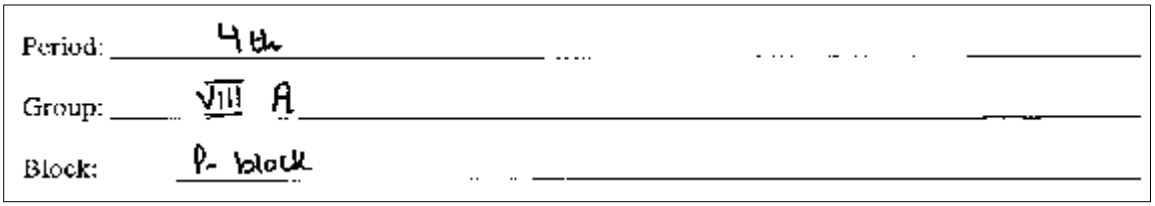
Question 1a:

Identify the period, group and block of an element based on its electronic configuration given below.

$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6$

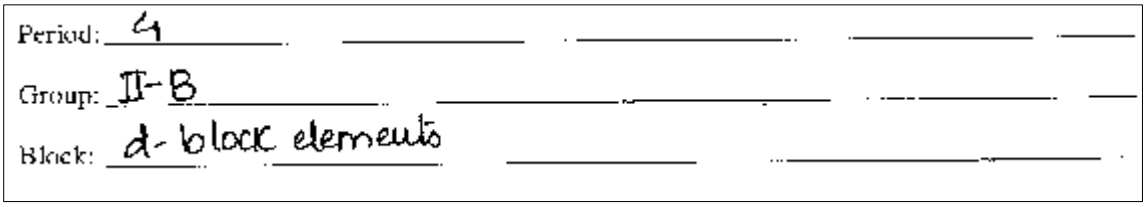
Better responses recognised the given electronic configuration to extract period, group and block of respective element. Candidates mentioned that the element belongs to period IV, group VIII A and it is from p-block of periodic table.

Example:



Weaker responses showed candidates' poor understanding of the valence shell electronic configuration. In most cases, candidates were not able to correctly identify the period, group and block of element through the given electronic configuration. However, some candidates correctly recognised the period as IV, but wrongly identified the group and block of element. These responses identified that the element belonged to d-block possibly due to the $4s^2$, $3d^{10}$ part of given electronic configuration.

Example:



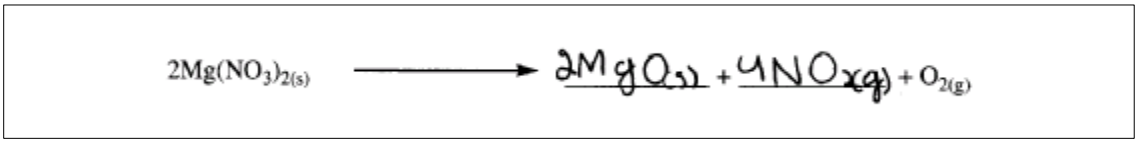
Question 1b:

Write the decomposition products of the given nitrate of magnesium (Mg).

$2Mg(NO_3)_{2(s)} \longrightarrow$ _____ + _____ + $O_{2(g)}$

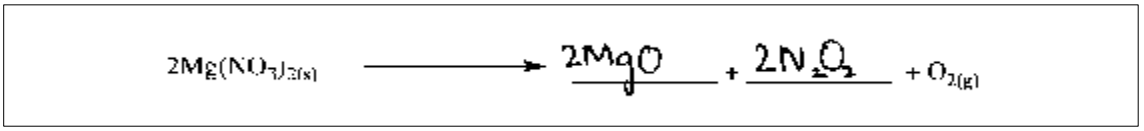
Better responses demonstrated accurate knowledge of the thermal stability of nitrates of group 2 elements. These responses mentioned both the missing decomposition products of the given nitrate of magnesium as MgO and NO_2, and balanced the given equation with correct numbers.

Example:



Weaker responses revealed that candidates wrote only one missing product, and were unable to identify the second product. Few of these responses showed both products but were unable to balance the given equation.

Example:



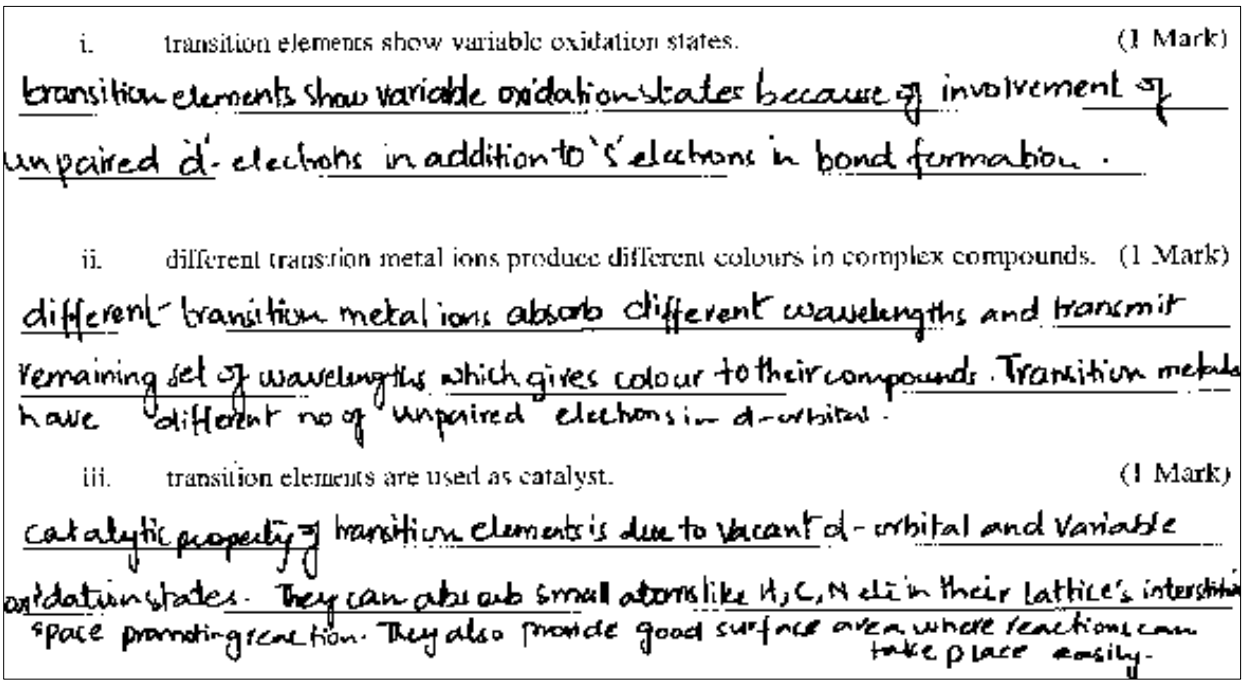
Question 2a:

Give suitable reasons why

i. transition elements show variable oxidation states.
ii. different transition metal ions produce different colours in complex compounds.
iii. transition elements are used as catalyst.

Better responses provided suitable reason in relation to transition elements for their variable oxidation state, colour and application as catalyst. In part i, candidates mentioned that transition elements possess unpaired d electrons due to which they show variable oxidation state. In part ii, they discussed that different transition metal ion absorb a different wavelength and transmit the remaining set of wavelengths that gives different colours to the ions. In part iii, it was described that transition elements show variety of oxidation states which allow them to form intermediate products and used as catalyst.

Example:



i. transition elements show variable oxidation states. (1 Mark)

transition elements show variable oxidation states because of involvement of unpaired d- electrons in addition to 's' electrons in bond formation.

ii. different transition metal ions produce different colours in complex compounds. (1 Mark)

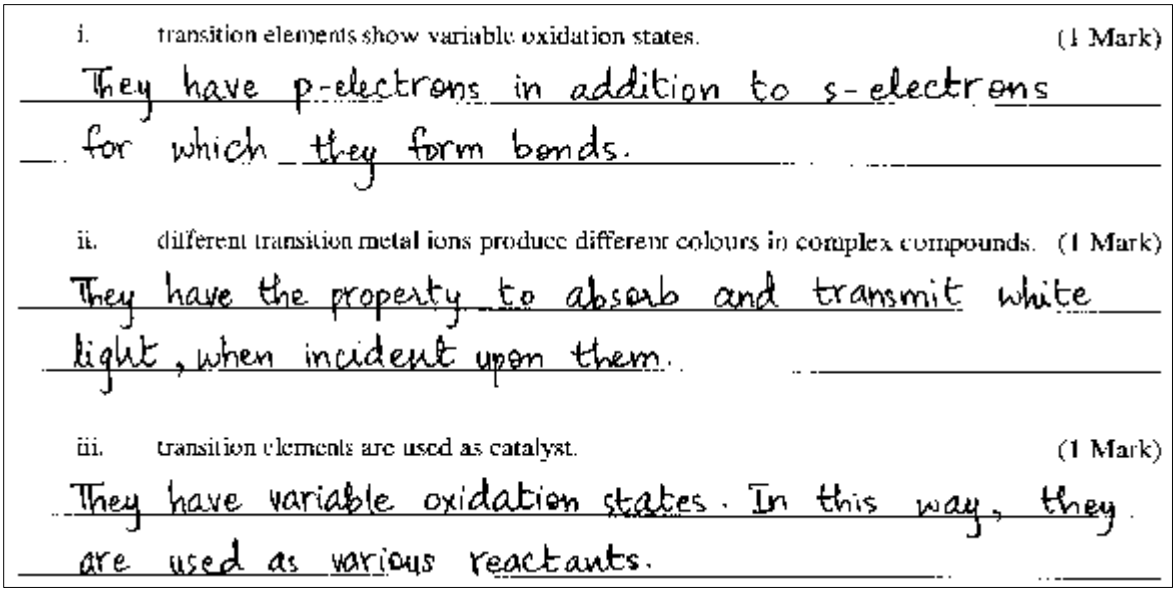
different transition metal ions absorb different wavelengths and transmit remaining set of wavelengths which gives colour to their compounds. Transition metals have different no of unpaired electrons in d-orbital.

iii. transition elements are used as catalyst. (1 Mark)

catalytic property of transition elements is due to vacant d- orbital and variable oxidation states. They can also absorb small atoms like H, C, N etc in their lattice's interstitial space promoting reaction. They also provide good surface area where reactions can take place easily.

Weaker responses gave irrelevant explanation for the required characteristics of transition elements. In part i, they discussed about isotopic nature of transition elements, partial filling of d orbitals, and presence of p electrons as responsible for variable oxidation state of transition elements. In part ii, instead of discussing energy difference of d-orbitals which varies from ion to ion, so they absorb a different wavelength light, these responses generally discussed the ability of transition elements to absorb and transmit incident light, transition of electron from lower energy level to higher energy level or vice versa. In part iii, candidates provided general explanation or the function of catalyst and mentioned that transition elements are used as catalyst because they can speed up the reaction. However, few of these responses mentioned about the variety of oxidation states in transition elements.

Example:



i. transition elements show variable oxidation states. (1 Mark)

They have p-electrons in addition to s-electrons for which they form bonds.

ii. different transition metal ions produce different colours in complex compounds. (1 Mark)

They have the property to absorb and transmit white light, when incident upon them.

iii. transition elements are used as catalyst. (1 Mark)

They have variable oxidation states. In this way, they are used as various reactants.

Question 2b:

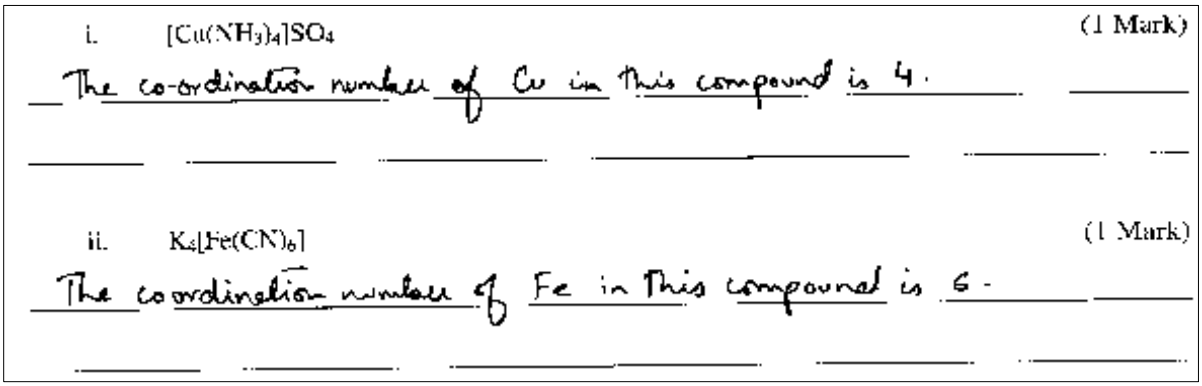
Identify the coordination number of Cu and Fe in the given coordination compounds.

i. $[Cu(NH_3)_4]SO_4$

ii. $K_4[Fe(CN)_6]$

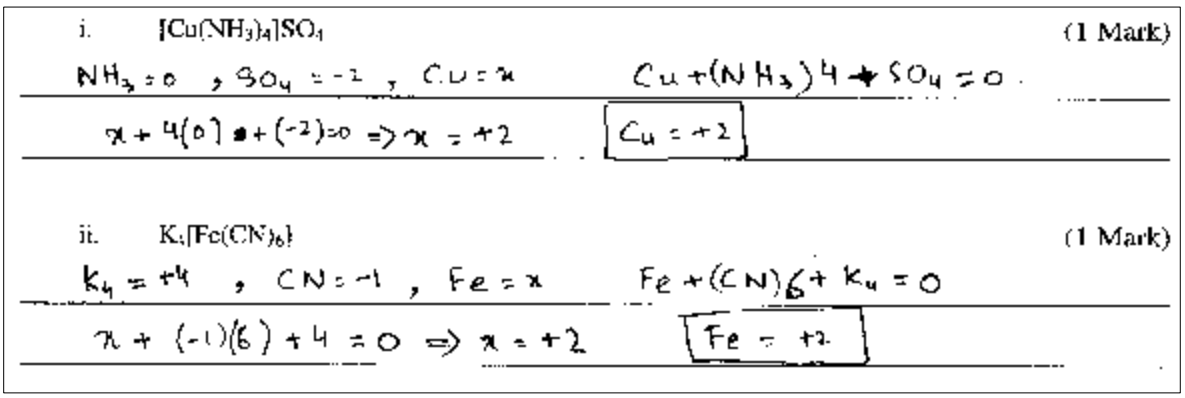
Better responses showed that candidates had a good understanding of ligand attached to central metal atom in coordination compounds, and correctly identified the coordination number of Cu and Fe as 4 and 6 respectively.

Example:

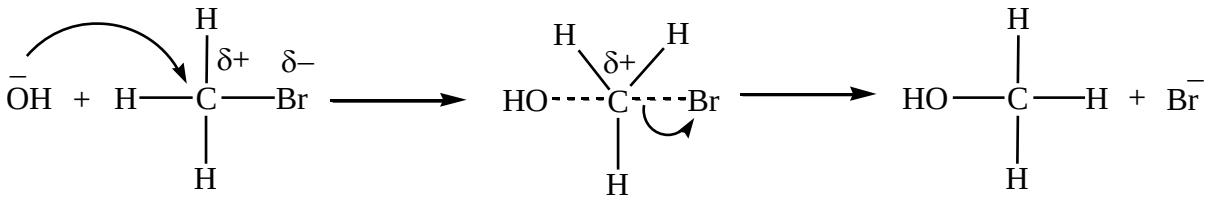


Weaker responses displayed wrong coordination number. In most of these responses, candidates tried to calculate the oxidation number of Cu and Fe instead of looking at the number and types of ligands attached to the central metal atom. Hence, these responses did not mention correct coordination number of Cu and Fe.

Example:



Question 3:



Transition state

Identify the following in the given mechanism of nucleophilic substitution (SN) reaction.

a. Type of nucleophilic substitution reaction
b. Type of alkyl halide (R–X) used
c. Leaving group
d. Number of molecules involved in the rate determining step

Better responses exhibited that candidates had an excellent understanding of the mechanism of nucleophilic substitution reactions. Candidates identified the type of nucleophilic substitution reaction as SN^2 and the type of alkyl halide correctly. Next, they recognised the leaving group and mentioned it as bromide ion or Br^-. Finally, they indicated that two molecules are involved in the rate determining step.

Example:

a. Type of nucleophilic substitution reaction (1 Mark)

SN2

b. Type of alkyl halide (R–X) used (1 Mark)

Primary alkyl halide (°1)

c. Leaving group (1 Mark)

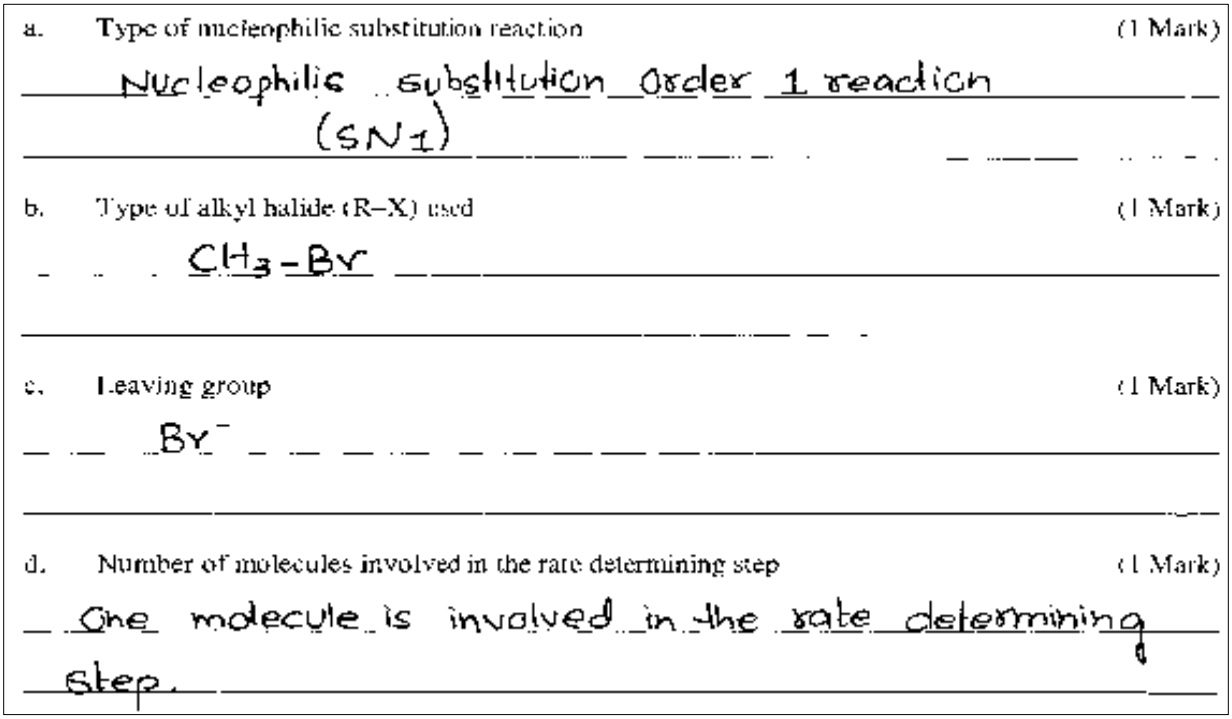
Br⁻

d. Number of molecules involved in the rate determining step (1 Mark)

2

Weaker responses failed to interpret the given mechanism clearly. They depicted poor concept of the type and mechanism of nucleophilic substitution reactions. First, they wrongly identified the type of nucleophilic substitution reaction as unimolecular or SN^1. Furthermore, instead of identifying the correct type of alkyl halide, they reproduced the same structure of alkyl halide that was already given in the mechanism. On the other hand, rather than mentioning the leaving group from the given mechanism, they identified bromine as a leaving group. Some of these responses wrote Br_2. These responses showed similar mistakes in part 'd' as in part 'a' and gave wrong number of molecules for the rate determining step. Because they identified the mechanism as SN^1 so they suggested that there is only one molecule involved in the rate determining step. However, in few of these responses, candidates were only able to perform well in part 'a' and 'c', but showed mistakes discussed in part 'b' and 'd'.

Example:


a. Type of nucleophilic substitution reaction (1 Mark)

Nucleophilic substitution order 1 reaction (SN1)

b. Type of alkyl halide (R–X) used (1 Mark)

CH_3-Br

c. Leaving group (1 Mark)

Br^-

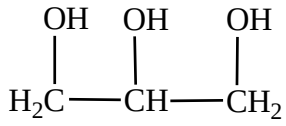
d. Number of molecules involved in the rate determining step (1 Mark)

One molecule is involved in the rate determining step.

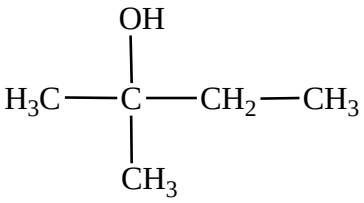
Question 4a:

Write the IUPAC names of the following alcohols.

i.



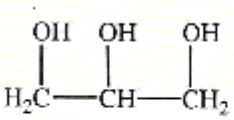
ii.



Better responses correctly identified the IUPAC name of the given alcohols as 1,2,3-Propanetriol and 2-Methyl-2-butanol in part i and ii respectively.

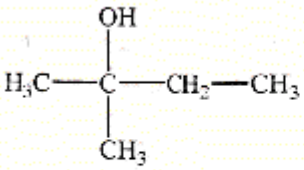
Example:

i. (1 Mark)



1, 2, 3- propantriol

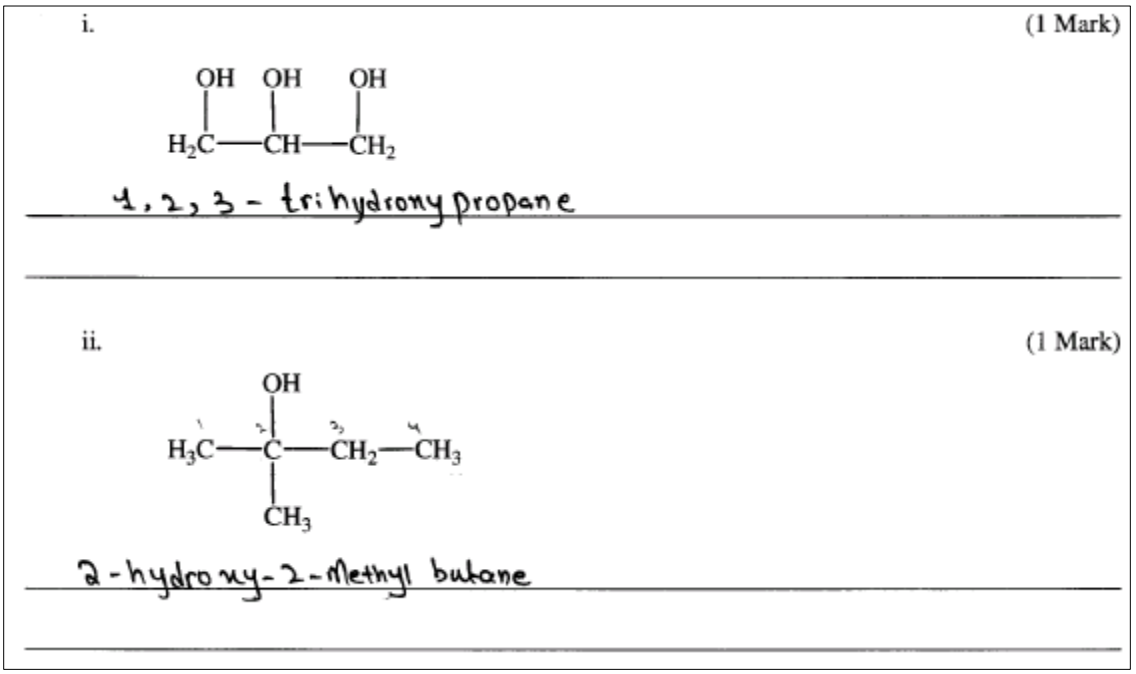
ii. (1 Mark)



2-Methyl-2-butanol

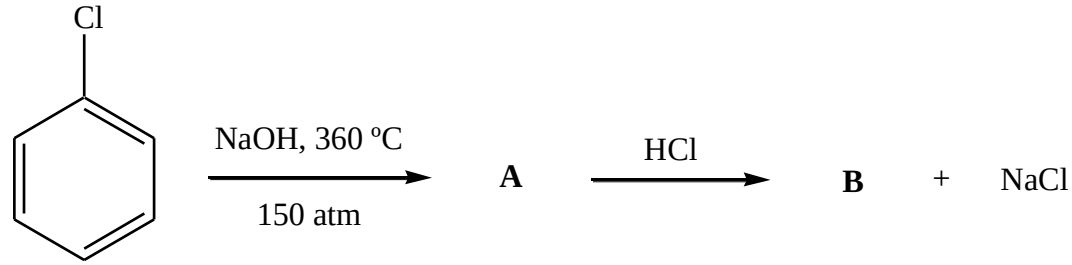
Weaker responses exhibited lack of knowledge about the IUPAC rules and depicted wrong names of the given alcohols. In part i, they selected the long chain of carbon for correct numbering, but gave incorrect names as 1,2,3-trihydroxypropane. Similarly, in part ii they were able to recognise the long chain of carbon with correct numbering, but did not write the correct IUPAC name.

Example:



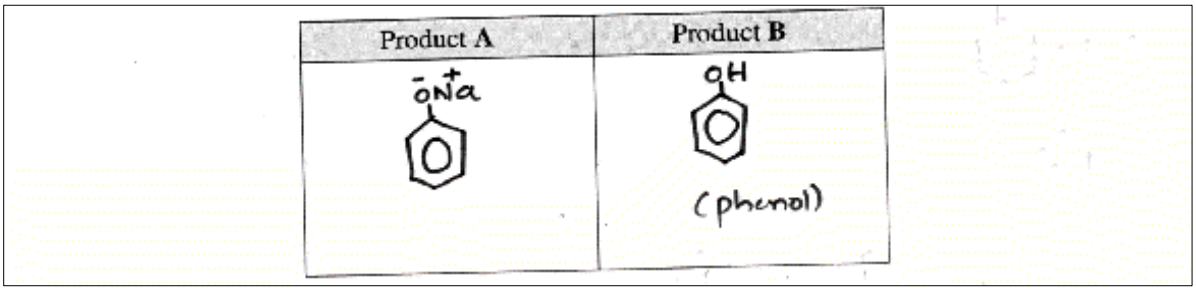
Question 4b:

Complete the given chemical reaction by drawing the structures of products **A** and **B**.



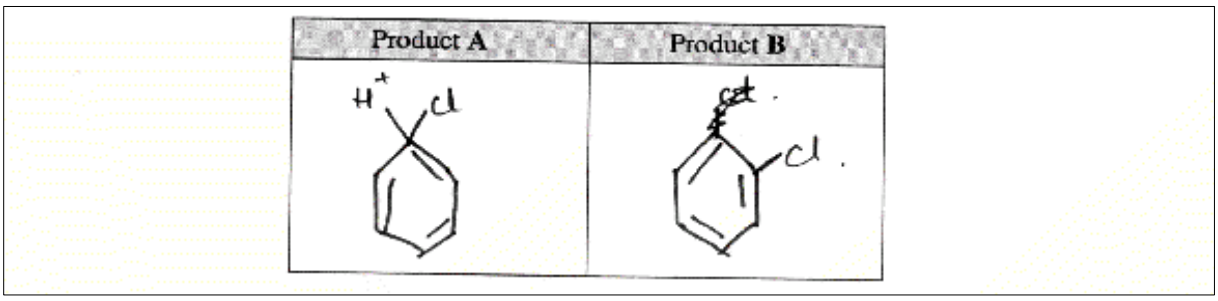
Better responses demonstrated good understanding of chemical reactions involving preparation of phenols. Candidates recognised the given reactant, reagent, and conditions very well including the sequence of reaction and formation of NaCl as side product. They drew the correct structure of products **A** (Sodium phenoxide) and **B** (Phenol) in appropriate boxes to complete the reaction.

Example:



Weaker responses were unable to identify each product in the given equation and draw wrong structure of product **A** and **B**. They showed poor knowledge about the reactions and preparation of phenol. Some of these responses were able to identify the product **A** and **B**, but did not show aromaticity. Similarly, some of them identified the product correctly, but replaced the product **A** with **B** and vice versa. Few of these responses wrote the name of products **A** and **B** rather than drawing a structure of products as asked in the question.

Example:

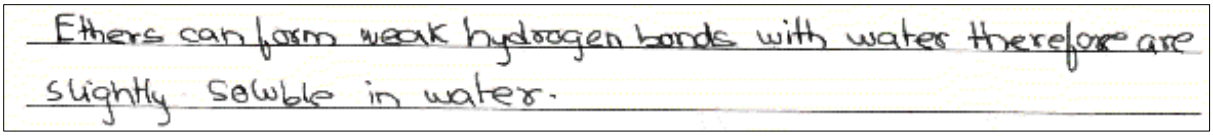


Question 4c:

Why are ethers slightly soluble in water?

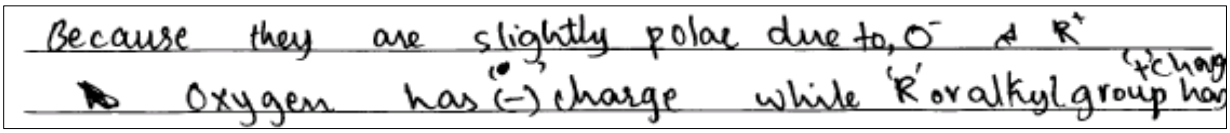
Better responses provided the correct answer in relation to the solubility of ethers in water. Candidates mentioned that ethers are slightly soluble in water because of weak hydrogen bonding with water molecules.

Example:



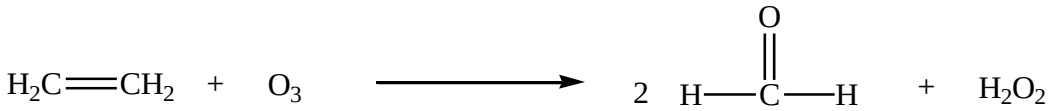
Weaker responses tried to discuss about the polarity, charges or intermolecular forces in ether molecule, but did not discuss about the weak hydrogen bonding that exists between ether and water molecule, so they were unable to answer correctly.

Example:



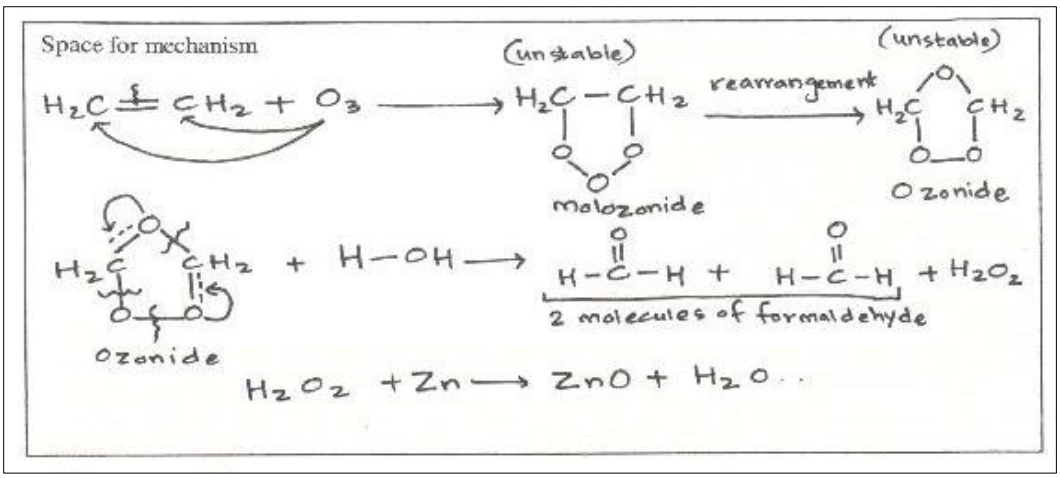
Question 5a:

Write the mechanism for the given reactions of ethene.



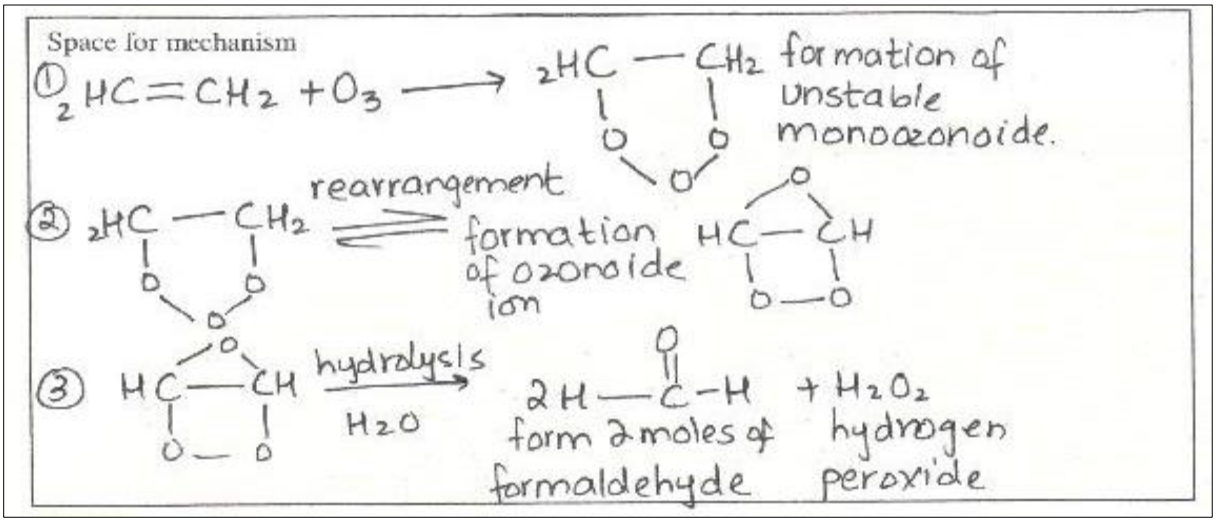
Better responses correctly displayed key steps involved in the mechanism for the preparation of two molecules of formaldehyde by the reaction of ethene with ozone. Candidates showed correct structure of molozonide, rearrangement of molozonide to ozonide and then conversion into two molecules of formaldehyde.

Example:



Weaker responses showed that candidates struggled to display important steps of mechanism involved in the preparation of formaldehyde. In most of these responses, candidates drew the wrong structure of molozonide and ozonide and showed inappropriate connectivity of C-C, C-O and C-H bonds. Some of these responses were able to draw the structure of molozonide correctly, but were unable to show the rearrangement in the structure to form ozonide.

Example:



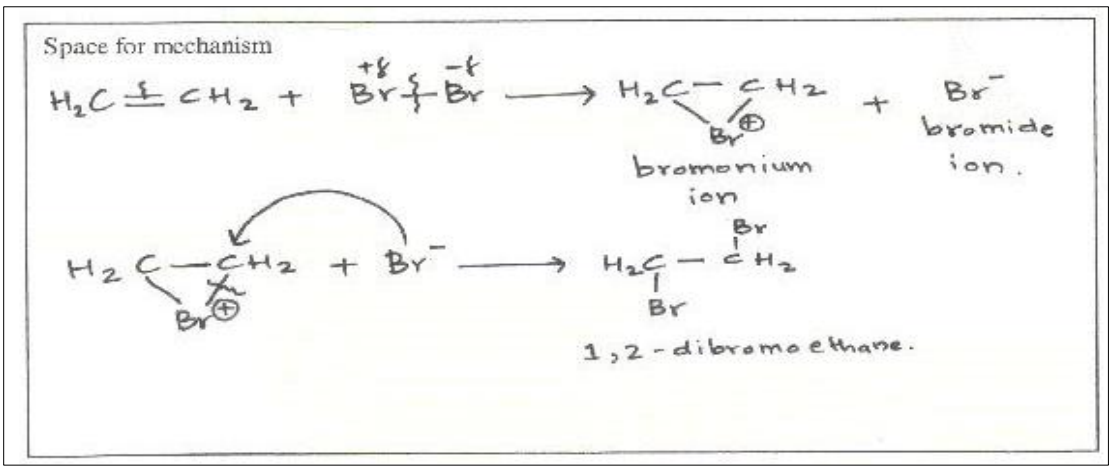
Question 5b:

Write the mechanism for the given reactions of ethene.



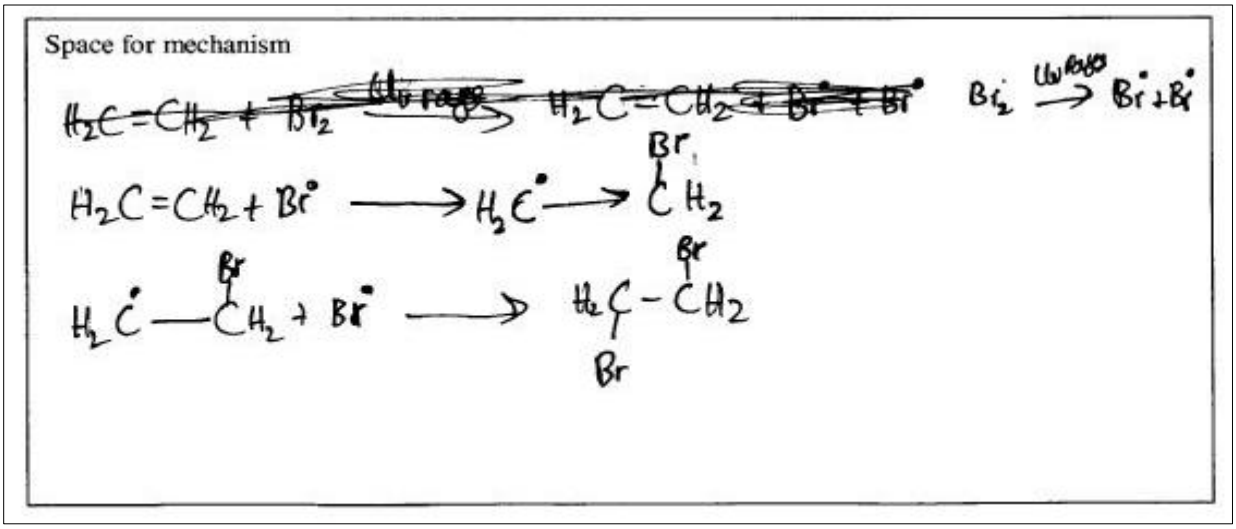
Better responses depicted in-depth understanding of addition reaction of alkene and nicely constructed the required mechanism for the given reaction. They showed the formation of cyclic bromonium ion intermediate clearly, and then the attack of bromide ion on cyclic intermediate to form addition product.

Example:



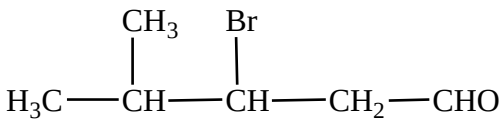
Weaker responses displayed that candidates misunderstood the given reaction and hence wrote wrong mechanism. These responses considered the given reaction as free radical halogenation reaction, and tried to frame its mechanism. Some of these responses were able to identify it as addition reaction of ethene, but showed the structure of carbocation instead of drawing the structure of cyclic bromonium ion intermediate.

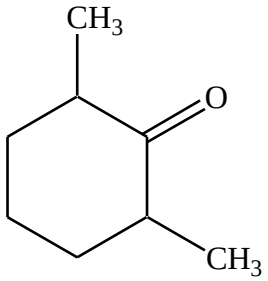
Example:



Question 6a

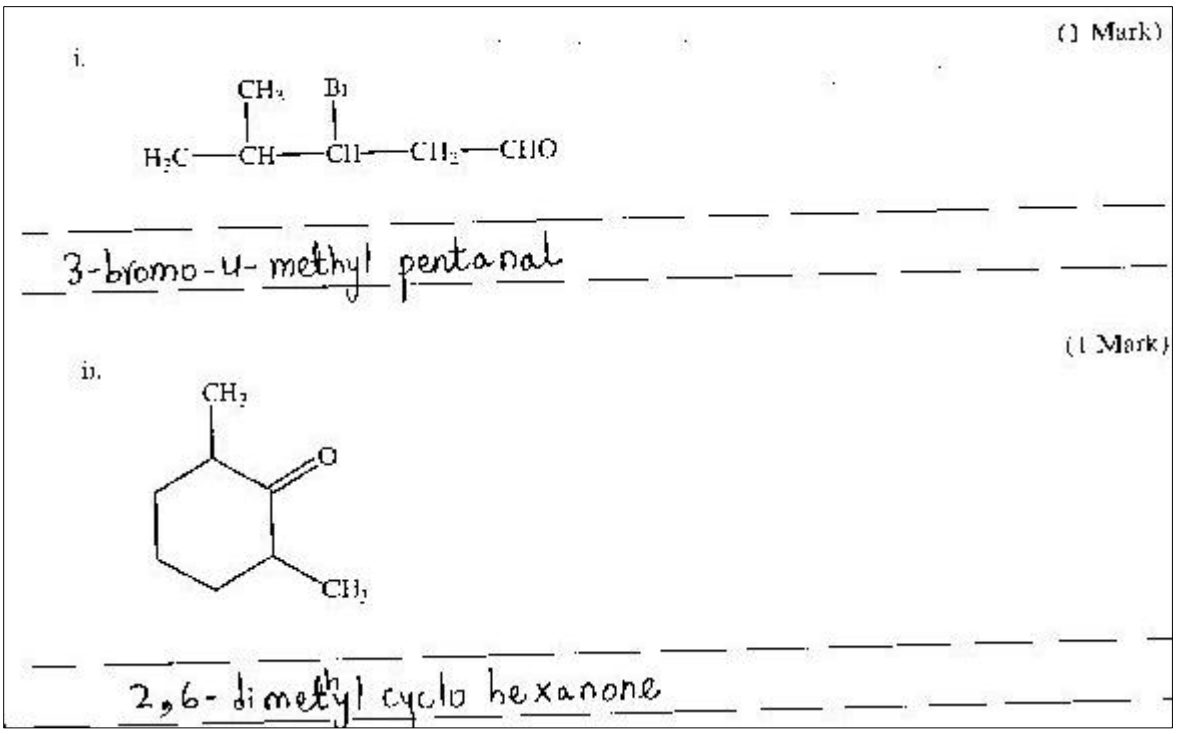
Write the IUPAC names of the following carbonyl compounds.

i. 

ii. 

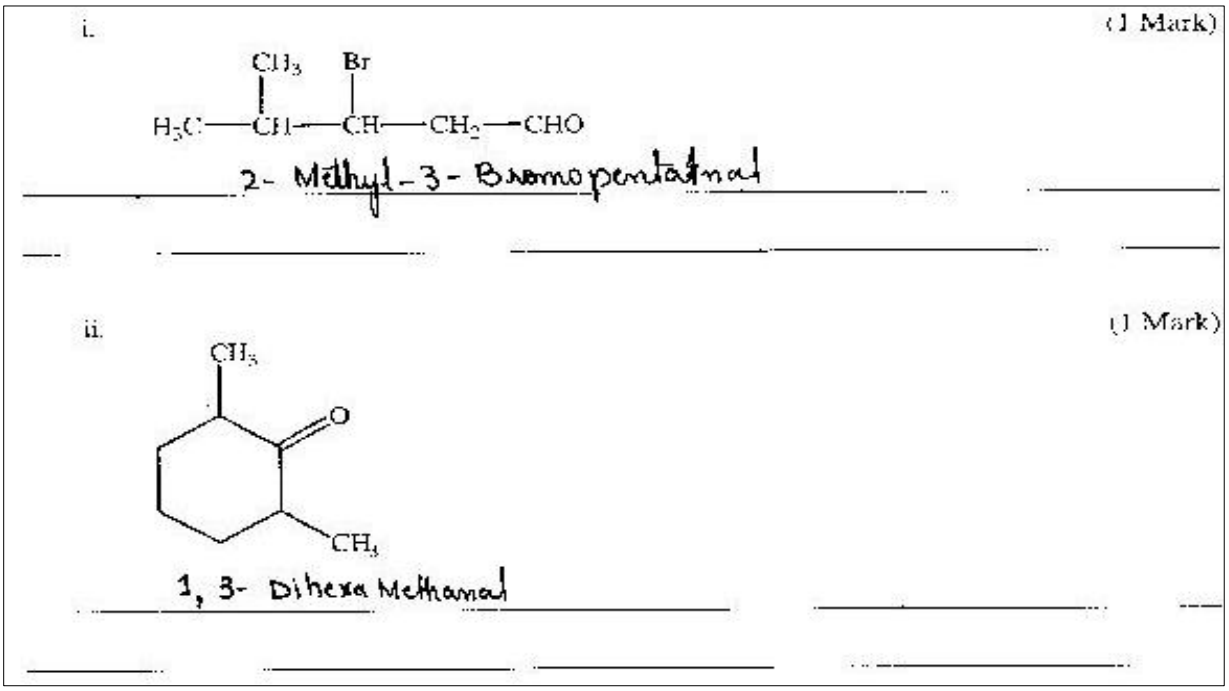
Better responses were able to identify the IUPAC name of the given carbonyl compound with correct numbering and functional groups present. They mentioned the name of given carbonyl compounds as 3-bromo-4-methyl pentanal and 2,6-dimethyl cyclohexanone in part i and ii respectively.

Example:



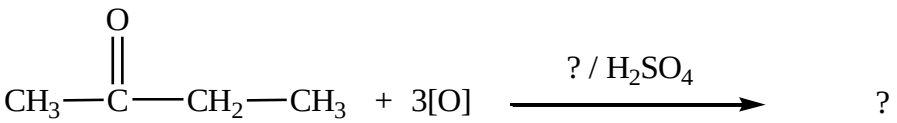
Weaker responses showed careless mistakes in writing the IUPAC name of given carbonyl compounds. Candidates displayed wrong numbering in carbon chain or wrong name. In part i, these responses started numbering from the terminal methyl group rather than starting from the carbonyl carbon atom and depicted the name of given compound as 2-methyl-3-bromopentanal. Some of these responses mentioned the position of halogen correctly, but identified methyl group at wrong position such as 3-bromo-2-methyl pentanal. In part ii, these responses wrote hexanone instead of cyclohexanone or recognised methyl group at wrong position such as at 1 and 3 rather than 2 and 6.

Example:



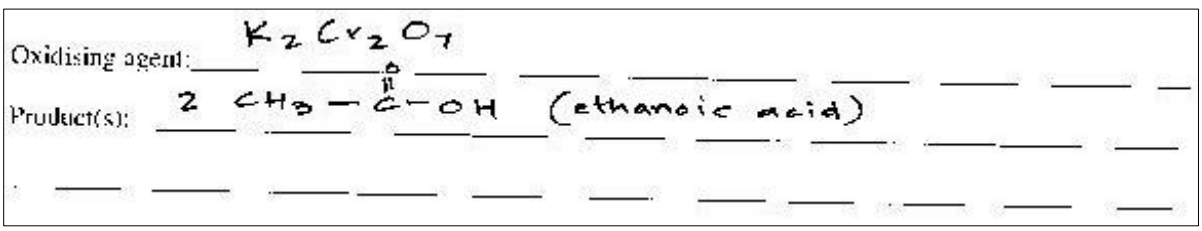
Question 6b

Identify the oxidising agent and the product(s) in the given chemical reaction.



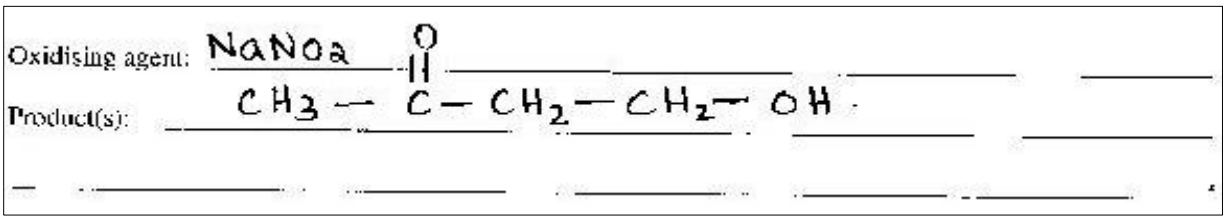
Better responses identified the correct oxidizing agent used and the product formed as a result of oxidation of given compound. They mentioned $K_2Cr_2O_7$ as an oxidising agent and represented two molecules of acetic acid (CH_3COOH) as product.

Example:



Weaker responses showed candidates' poor understanding of the oxidation reactions of carbonyl compounds like aldehydes and ketones. These responses were not able to correctly identify the oxidizing agent used in the given reaction and the product formed as a result of oxidation of given ketone. Instead of mentioning $K_2Cr_2O_7$ as an oxidising agent they showed variety of other reagents irrelevant to the given reaction such as $NaNO_2$. Similarly, they were unable to mention correct structure of product.

Example:



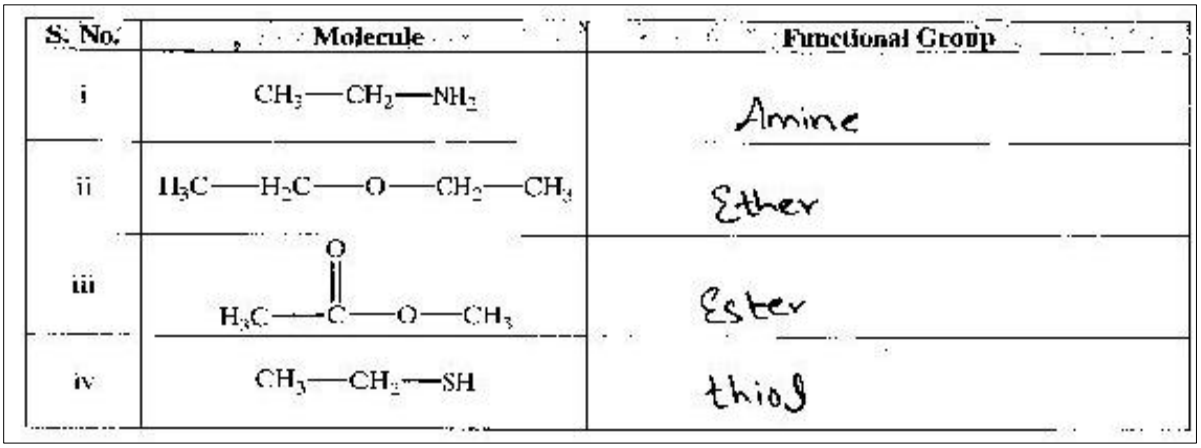
Question 7:

Identify the functional group present in each of the given molecules.

S. No.	Molecule	Functional Group
i	$CH_3—CH_2—NH_2$	
ii	$H_3C—H_2C—O—CH_2—CH_3$	
iii	$H_3C—\overset{\overset{O}{\|\|}}{C}—O—CH_3$	
iv	$CH_3—CH_2—SH$	

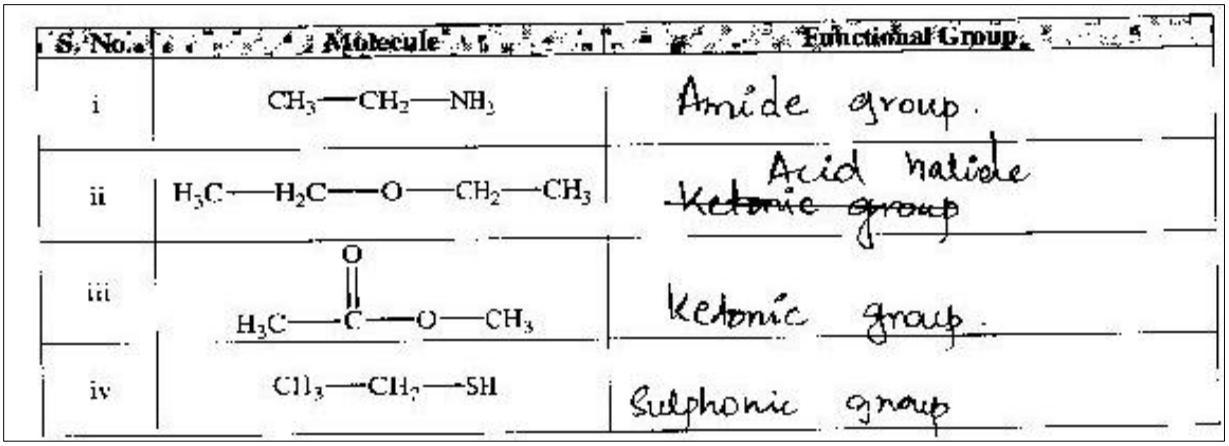
Better responses demonstrated that candidates had good knowledge about the different types of functional group. Candidates easily identified the functional group as amine, ether, ester and thiol in each given molecule.

Example:



Weaker responses failed to correctly highlight each type of functional group. Some of these responses mentioned amide as functional group instead of amine and sulfone instead of thiol. Apart from it few responses showed completely wrong judgement and identified ether as acid hlide and ester as ketone.

Example:

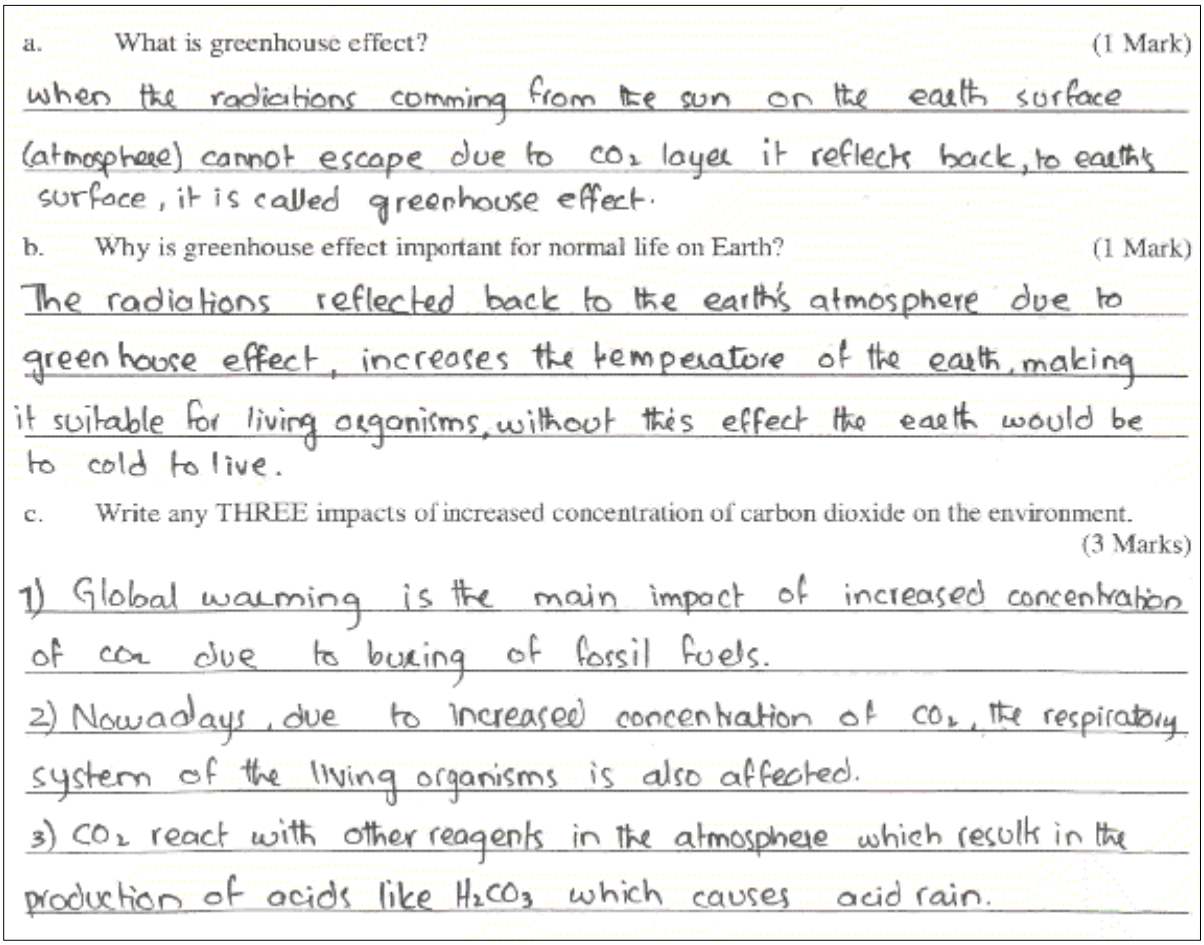


Question 8:

a. What is greenhouse effect?
b. Why is greenhouse effect important for normal life on Earth?
c. Write any THREE impacts of increased concentration of carbon dioxide on the environment.

Better responses explained greenhouse effect clearly using all the keywords like sun's energy reaches the Earth's atmosphere, some of it is reflected back to space and the rest is absorbed and re-radiated by greenhouse gases. In part b, candidates further mentioned that without the greenhouse effect life is not possible as the whole Earth would be converted into extremely cold planet. In part c, they described impacts of increased concentration of CO_2 in the environment, i.e. melting of glaciers, increased precipitation (rainfall and snowfall), increased allergy and asthma rates, more extreme weather etc.

Example:



a. What is greenhouse effect? (1 Mark)

when the radiations comming from the sun on the earth surface (atmosphere) cannot escape due to CO_2 layer it reflects back, to earth's surface, it is called greenhouse effect.

b. Why is greenhouse effect important for normal life on Earth? (1 Mark)

The radiations reflected back to the earth's atmosphere due to greenhouse effect, increases the temperature of the earth, making it suitable for living organisms, without this effect the earth would be to cold to live.

c. Write any THREE impacts of increased concentration of carbon dioxide on the environment. (3 Marks)

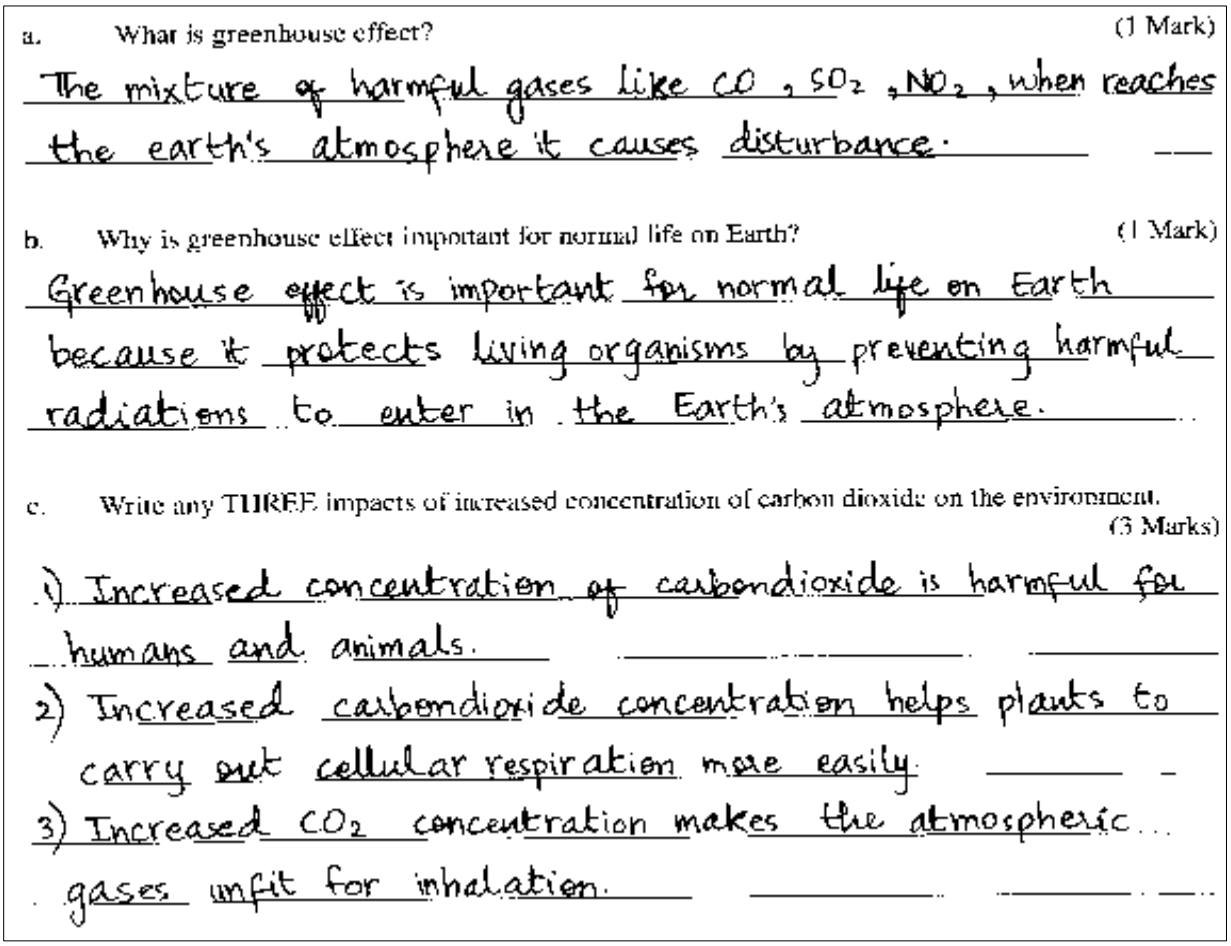
1) Global warming is the main impact of increased concentration of CO_2 due to burning of fossil fuels.

2) Nowadays, due to increased concentration of CO_2, the respiratory system of the living organisms is also affected.

3) CO_2 react with other reagents in the atmosphere which result in the production of acids like H_2CO_3 which causes acid rain.

Weaker responses demonstrated weak understanding of the concept of greenhouse effect. These responses were unable to provide correct definition of greenhouse effect and tried to give general explanation such as 'radiations from the sun absorb in the atmosphere and cause disturbance'. In part b, these responses described the importance of greenhouse effect wrongly and mentioned that it is necessary because it provides safety from the harmful radiation. Most of these responses performed little well in part c compared to part a and b. They were not able to provide more specific answer to the question, but mentioned few points generally related to increased concentration of CO_2 such as inhalation problem.

Example:



a. What is greenhouse effect? (1 Mark)

The mixture of harmful gases like CO , SO_2 , NO_2 , when reaches the earth's atmosphere it causes disturbance.

b. Why is greenhouse effect important for normal life on Earth? (1 Mark)

Greenhouse effect is important for normal life on Earth because it protects living organisms by preventing harmful radiations to enter in the Earth's atmosphere.

c. Write any THREE impacts of increased concentration of carbon dioxide on the environment. (3 Marks)

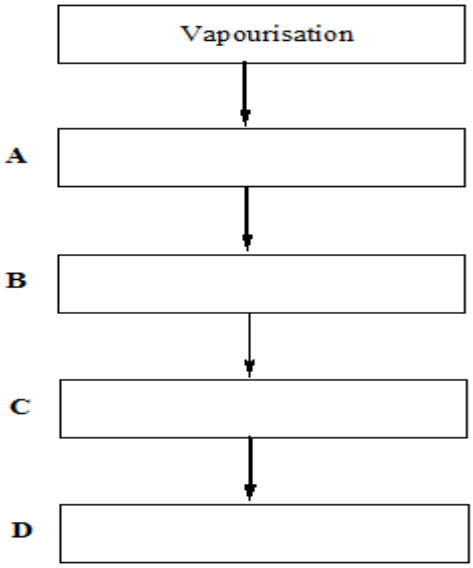
1) Increased concentration of carbondioxide is harmful for humans and animals.

2) Increased carbondioxide concentration helps plants to carry out cellular respiration more easily.

3) Increased CO_2 concentration makes the atmospheric gases unfit for inhalation.

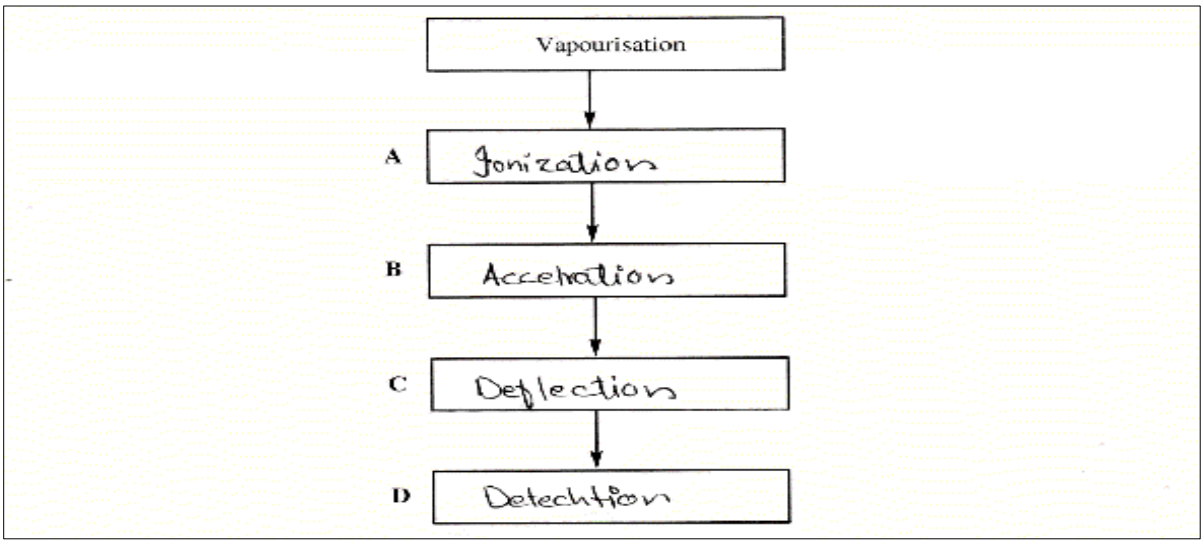
Question 9:

Name the four stages, **A** to **D**, of a mass spectrometer in sequence in the given boxes.



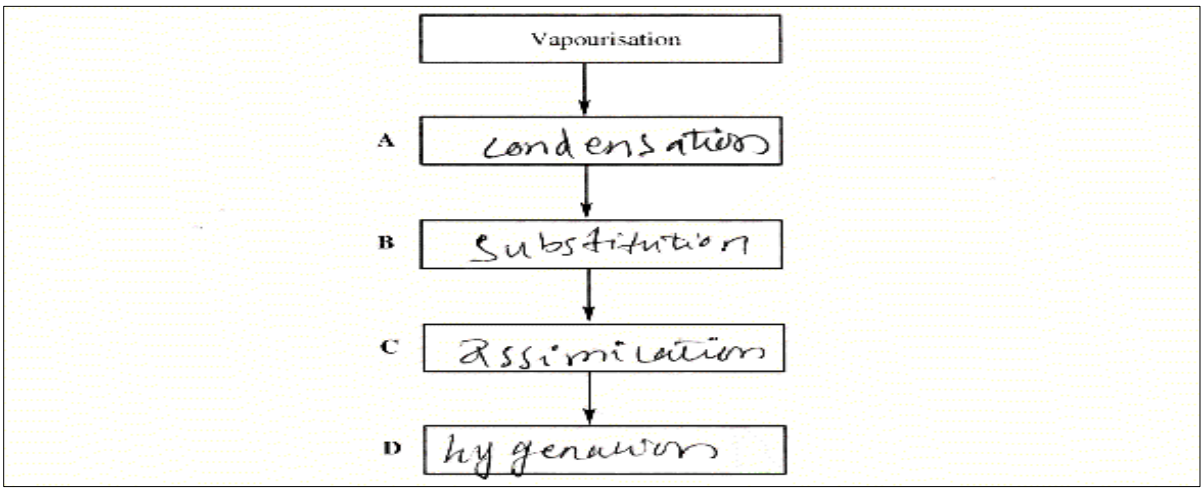
Better responses managed to spot each stage of the mass spectrometer correctly. The first stage after vapourisation was identified as stage **A** (ionisation) in which atom or molecule is ionised. Next is the stage **B** (acceleration) where ions are accelerated, and stage **C** (deflection) in which ions are deflected by a magnetic field according to their masses. Finally, the stage **D** (detection) where the beam of ions passes through the machine is detected electrically.

Example:



Weaker responses made the wrong judgment about each stage of the mass spectrometer. For each stage, candidates provided illogical and irrelevant names such as for stage **A** as condensation, stage **B** as substitution, stage **C** as assimilation, and stage **D** as hydrogenation. Some of these responses used other version of names for each stage like displacement, separation, settlement. However, few of these responses were only able to identify stage **A** correctly.

Example:



Extended Response Questions (ERQs)

These questions offered a choice between part **a** and **b**

Most of the candidates attempted **10a** and performed better compared to **10b**.

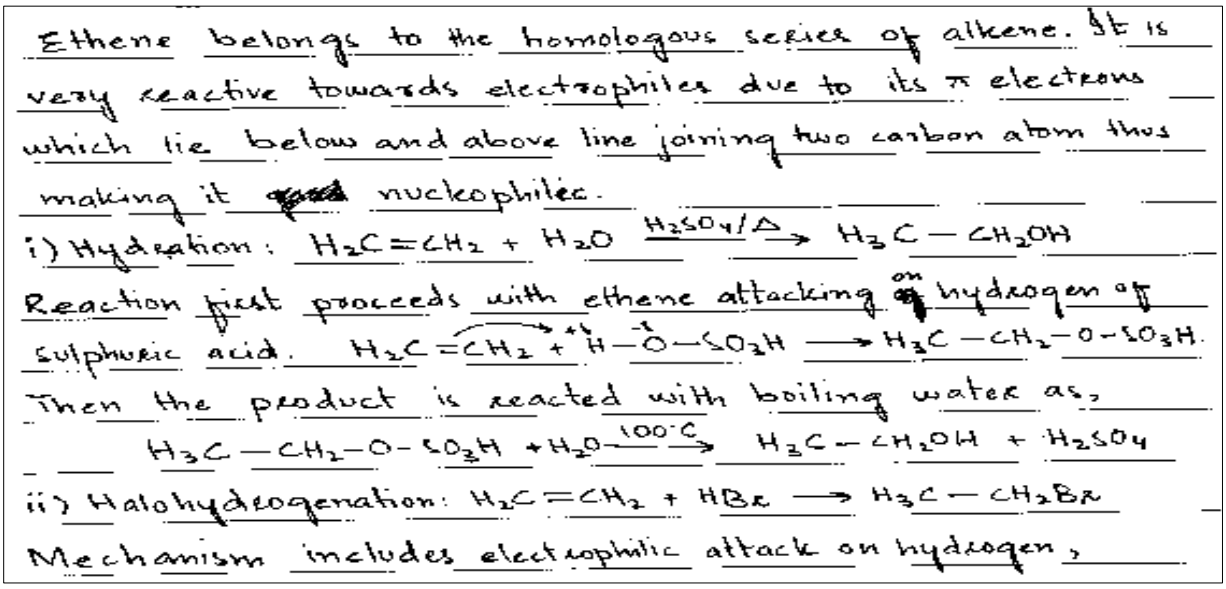
Question 10a

Describe the addition reactions of ethene with the following electrophilic reagents using balanced chemical equations. Mention conditions, i.e. catalyst, temperature and/ or pressure, where essential.

i. Sulphuric acid (hydration)
ii. Hydrogen bromide
iii. Hypochlorous acid
iv. Hydrogen (hydrogenation)

Better responses described the addition reaction of ethene with a variety of given reagents correctly using chemical equation. In part i, the reaction in which ethene is reacted with sulphuric acid, was identified as hydration of alkene. Candidates mentioned the correct condition for this reaction and described the formation of ethyl hydrogen phosphate which on hydrolysis yield ethanol as final product. In part ii, candidates mentioned addition of hydrogen bromide or hydrohalogenation, where in the first step they showed the formation of carbocation, which gets attacked by halide ion to give ethyl bromide as product. In part iii, they described the reaction as halohydration, in which ethene is reacted with hypochlorous acid. These responses mentioned the correct formula of hypochlorous acid and showed the formation of ethylene chlorohydrin or 2-chloroethanol. In part iv, they described the hydrogenation of alkene and showed the formation of ethane using chemical equation and by mentioning the appropriate conditions (catalyst and temperature).

Example:


Ethene belongs to the homologous series of alkene. It is very reactive towards electrophiles due to its π electrons which lie below and above line joining two carbon atom thus making it nuckophilec.

i) Hydration: $H_2C=CH_2 + H_2O \xrightarrow{H_2SO_4/\Delta} H_3C-CH_2OH$

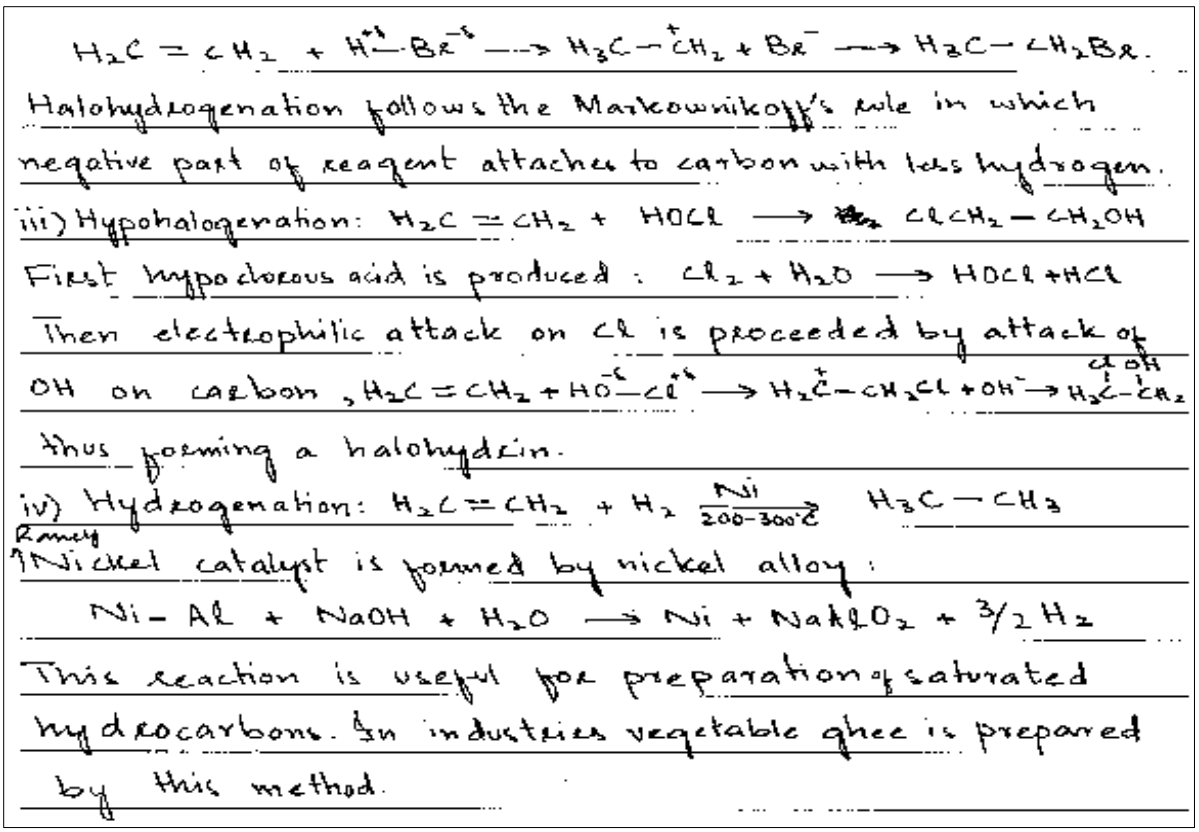
Reaction first proceeds with ethene attacking on hydrogen of sulphuric acid. $H_2C=CH_2 + {}^{+\delta}H-{}^{-\delta}O-SO_3H \longrightarrow H_3C-CH_2-O-SO_3H$

Then the product is reacted with boiling water as,

$H_3C-CH_2-O-SO_3H + H_2O \xrightarrow{100^\circ C} H_3C-CH_2OH + H_2SO_4$

ii) Halohydrogenation: $H_2C=CH_2 + HBr \longrightarrow H_3C-CH_2Br$

Mechanism includes electrophilic attack on hydrogen,



Weaker responses exhibited a number of mistakes while mentioning the reactions of alkene with different reagents. Candidates were unable to show correct structure of intermediates and final product. These responses did not mention the reaction conditions where essential. In part i, these responses did not indicate the correct condition of reaction. They were unable to show the formation of ethyl hydrogen phosphate in the first step. Additionally, instead of mentioning the structure of ethanol, candidates gave structure of propanol. Similarly, in part ii, the formation of carbocation intermediate was not shown. Candidates gave halogenation product in their final answer by showing the addition of Br_2 rather than indicating the structure of hydrohalogenation product (ethyl bromide) by addition of HBr. In part iii, most candidates were unable to identify the correct structure of hypochlorous acid, so they displayed reaction of ethene with HCl, and finally showed wrong product as chloroethane instead of 2-chloroethanol. In part iv, these responses failed to display the actual reaction conditions such as catalyst and temperature required and depicted completely wrong structure of product.

Example:

① In addition reaction of ethene teagents are added and in this one H_2SO_4 is added with ethene.

$$CH_2=CH_2 + H_2SO_4 \xrightarrow[200\,atm]{H_2O} CH_3-CH_2-CH_2-OH + SO_2 + H_2O$$

② In the second equation HBr is added with ethene.

$$CH_2=CH_2 + HBr \xrightarrow[200\,atm]{H_2O} CH_3-\underset{\underset{Br}{|}}{\overset{\overset{H}{|}}{CH}}-\underset{\underset{H}{|}}{\overset{\overset{Br}{|}}{CH}}-CH_3$$

③ In the third equation Hypochloric acid is added with ethene.

$$CH_2=CH_2 + HCl \xrightarrow[450^\circ C]{LiAlH_4} CH_3-CH_2-Cl + H$$

④ In the 4th equation Hydrogen is added with ethene.

$$CH_2=CH_2 + H \xrightarrow[450^\circ C\,/200atm]{LiAlH_4/H_2O} CH_3-\overset{\overset{O}{\|}}{C}H-H$$

Question 10b

i. Write the steps of mechanism for the following reaction.

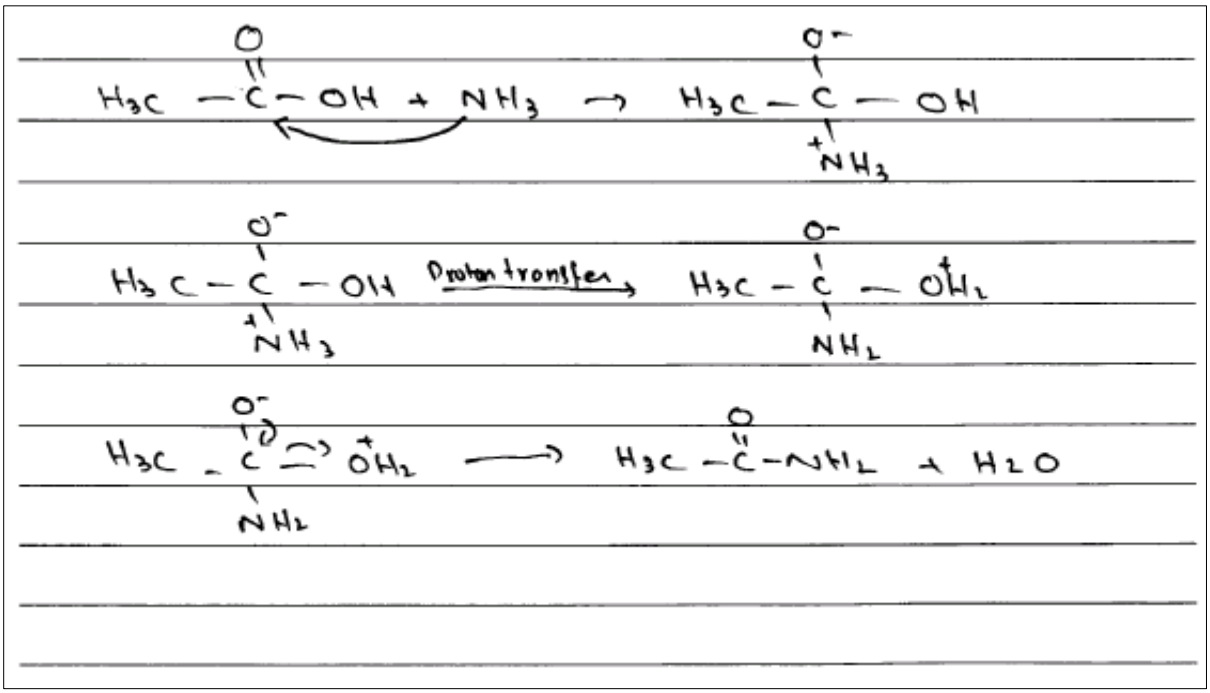


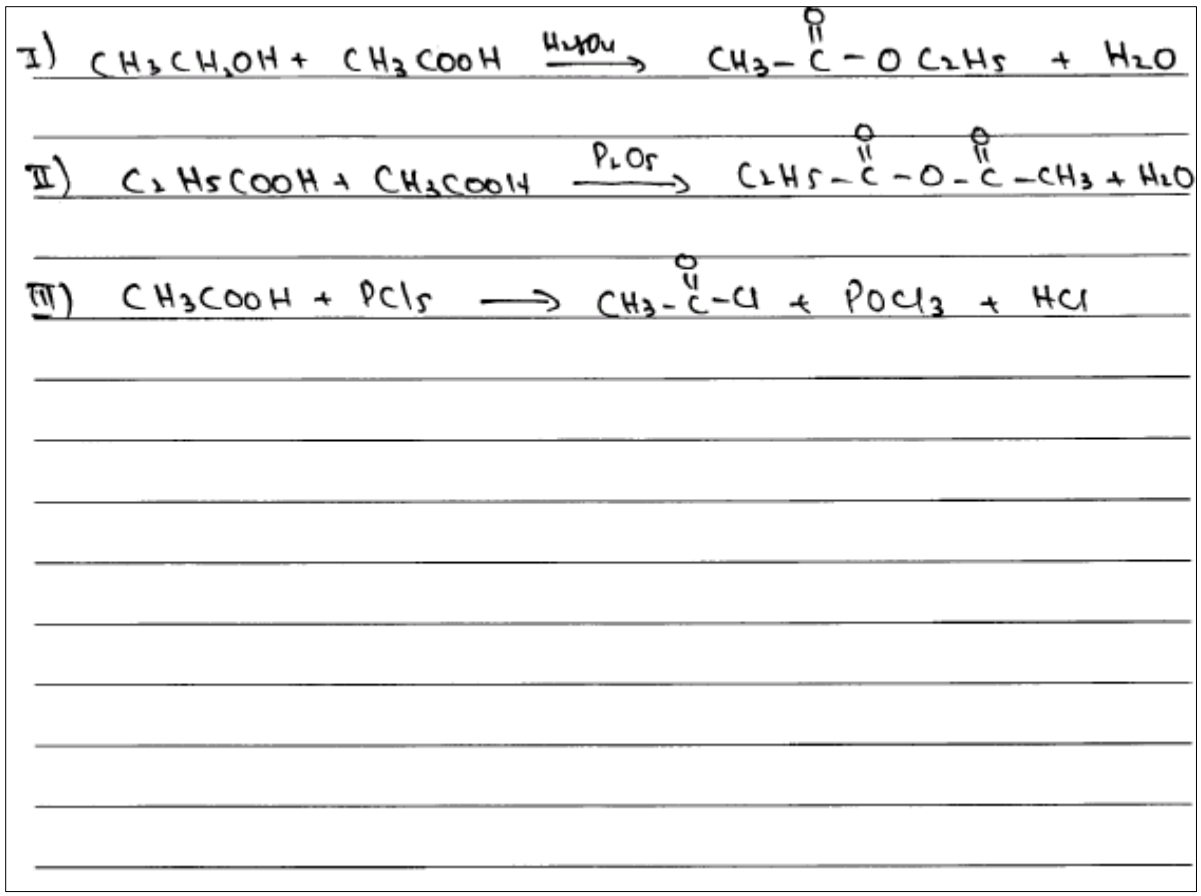
ii. Write chemical equations to show reaction of carboxylic acid (CH_3COOH) with the given reagents. Mention catalyst where required.

I. C_2H_5OH
II. CH_3CH_2COOH
III. PCl_5

Better responses showed that candidates had a good understanding of reactions involving preparation of acyl halides, acid anhydrides, esters and amides. In part i, they mentioned all the key steps involved in the conversion of acetic acid into amide. Candidates highlighted the attack of ammonia on carbonyl carbon of acetic acid to form tetrahedral intermediate. After that, the proton transfer was shown in the second step and finally the removal of water molecule to yield the final product as amide. In part ii, they displayed all three reactions using chemical equations mentioning the catalyst where required. Candidates showed the formation of ester by the reaction of acetic acid with ethanol under acidic condition. Similarly, formation of acetic anhydride and acid halide was illustrated, when acetic acid was reacted with another carboxylic acid and phosphorous pentachloride (PCl_5) respectively.

Example:



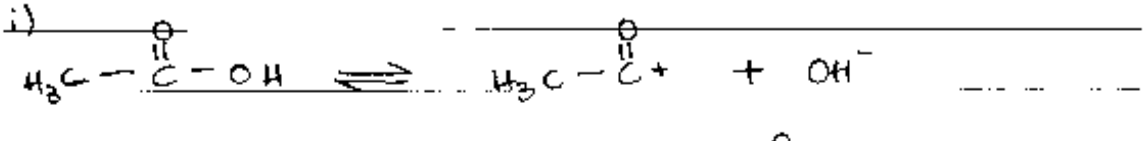


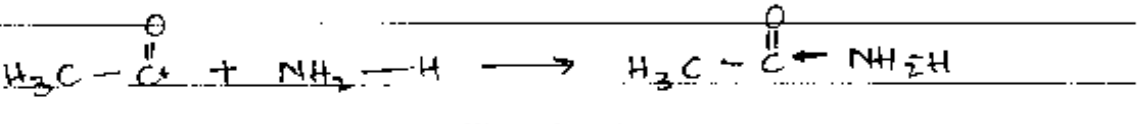
Weaker responses were unable to illustrate all the different steps involved in the mechanism of amide formation from acetic acid. Instead of mentioning the attack of ammonia on to carbonyl carbon to form tetrahedral intermediate, these responses directly displayed the formation of acylium cation, then they added ammonia to acylium cation and showed the removal of water molecule to furnish final product as amide. In part ii, most candidates were able to identify the product as ethyl acetate, but did not show the correct type of catalyst. These responses were not able to write correct structure of anhydride when two carboxylic acid molecules are reacted together. Some candidates were able to recognise the correct structure of acid anhydride, however, they were still unable to mention the catalyst required for this reaction. Finally, they showed careless errors while identifying the structure of acid halide when the acetic acid is reacted with phosphorous pentachloride (PCl_5). However, few of these responses showed correct formation of acid halide and other side products of that reaction.

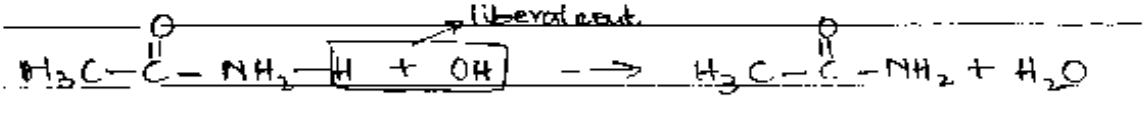
Example:

Option: B

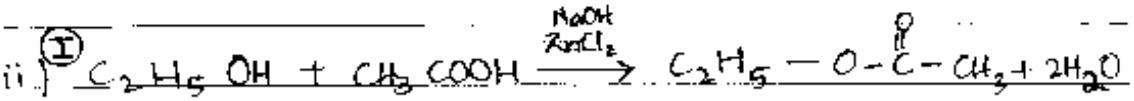
i)



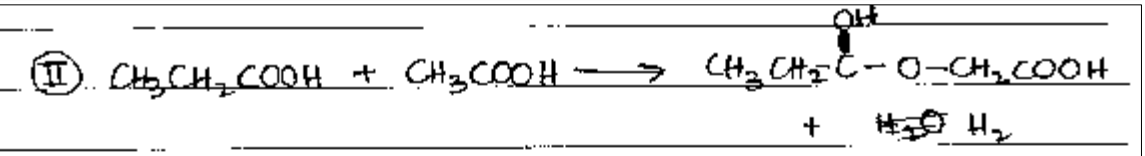




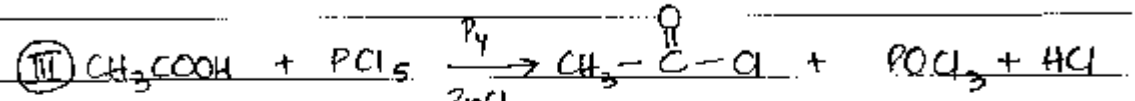
ii) (I)



(II)



(III)



Most of the candidates attempted **11b** and performed better compared to **11a**.

Question 11a

i. Write the role of each of the given pesticides.

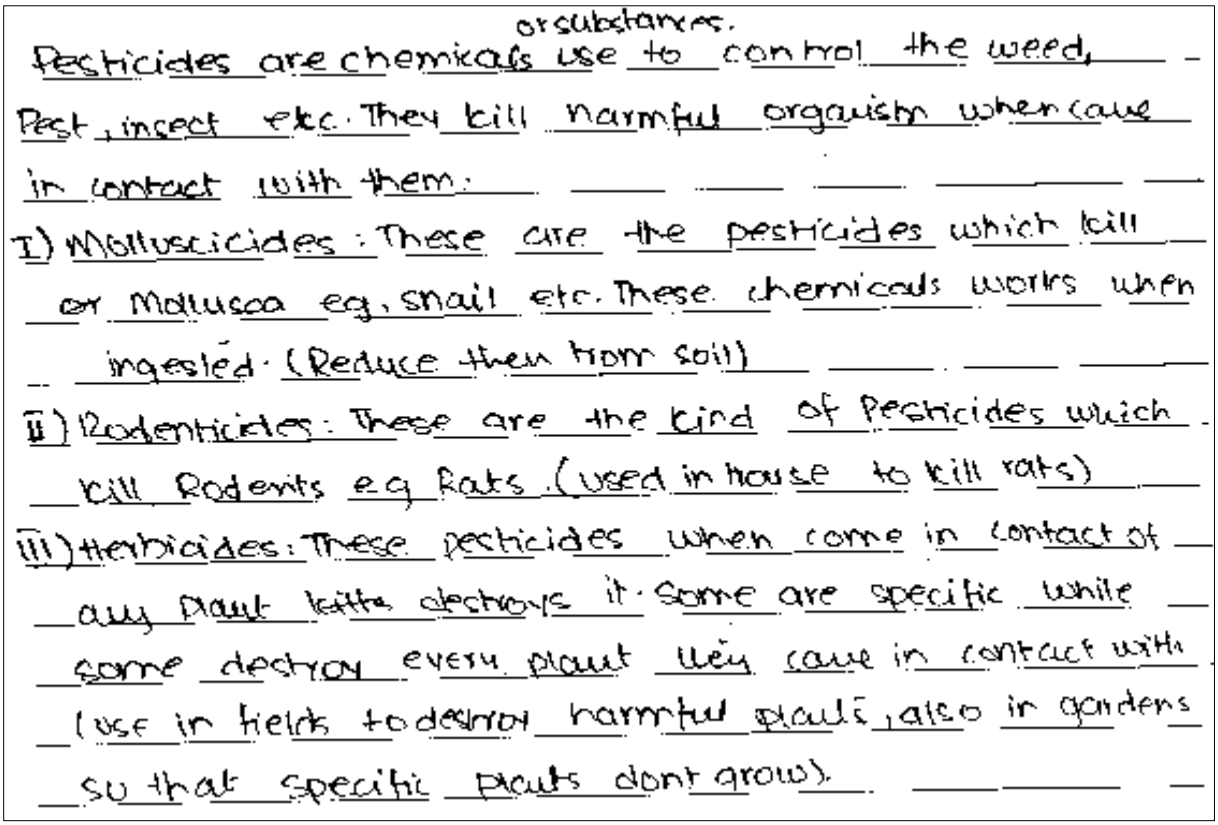
I. Molluscicides
II. Rodenticides
III. Herbicides
IV. Fungicides

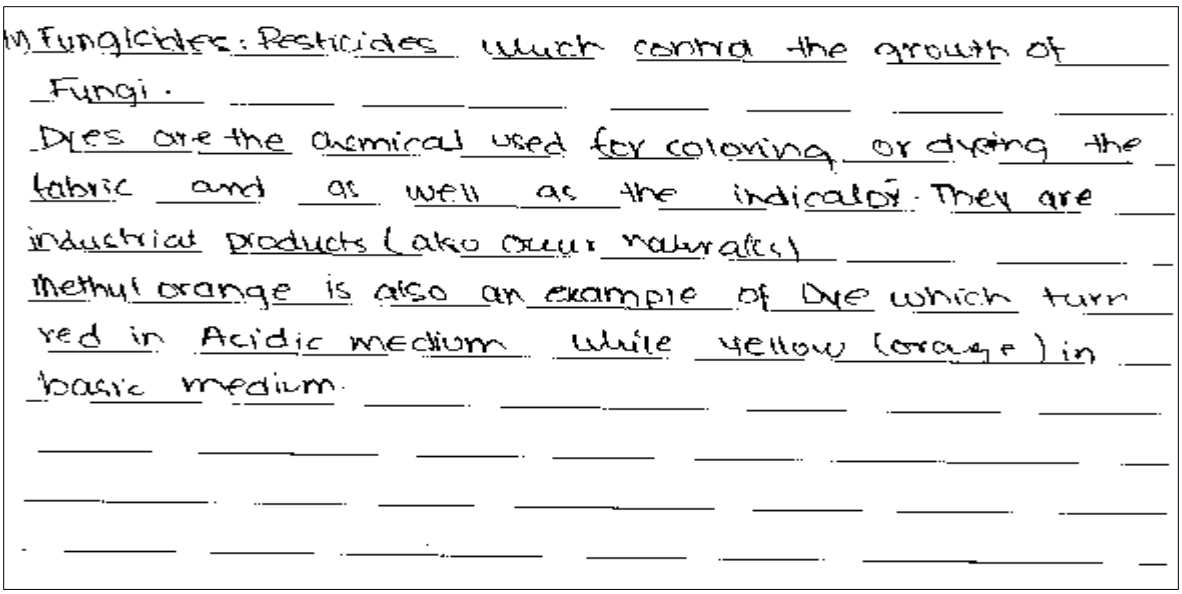
ii. What are dyes? What is the colour of methyl orange in

I. acidic medium?
II. basic medium?

Better responses demonstrated good knowledge of the concept of dyes and pesticides. Candidates described the role of each given pesticide correctly with some examples. They mentioned that molluscicides are chemicals used to control snails and slugs, and rodenticides are chemicals used to control rats, mice, bats and other rodents. Similarly, herbicides are used to control unwanted plants, and fungicides are chemicals used to control the fungi which cause molds, rots, and plant diseases. In part ii, they correctly defined dyes and identified the colour of methyl orange in acid and basic medium as red and yellow respectively.

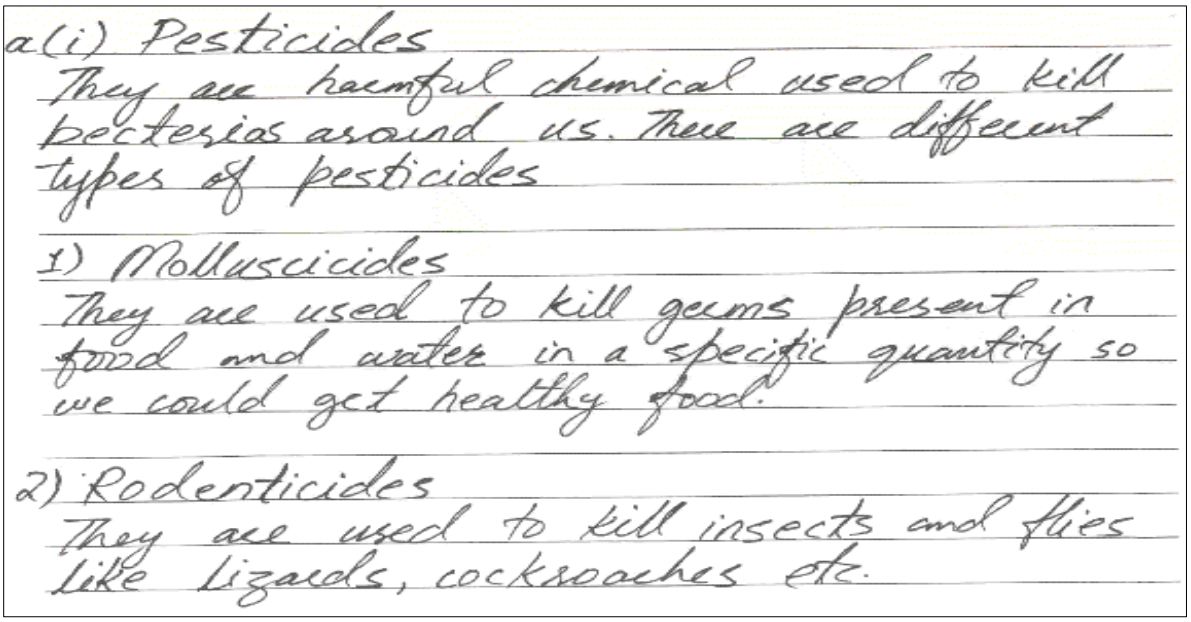
Example:


Pesticides are chemicals or substances use to control the weed, Pest, insect etc. They kill harmful orgaism when cave in contact with them:
I) Molluscicides: These are the pesticides which kill or Molusca eg, snail etc. These chemicals works when ingested. (Reduce them from soil)
II) Rodenticides: These are the kind of Pesticides which kill Rodents eg Rats. (used in house to kill rats)
III) Herbicides: These pesticides when come in contact of any plant kills destroys it. Some are specific while some destroy every plant they cave in contact with. (use in fields to destroy harmful plants, also in gardens so that specific plants dont grow).



Weaker responses provided incorrect explanation about dyes and their types. Candidates were not specific in their explanation and provided generalised answers. They described molluscicides as chemicals used to kill germs present in food and water. On the other hand, they illustrated that rodenticides are chemicals used to control cockroaches. These responses identified herbicides as chemicals used to save food from insects and bacteria. Moreover, it was mentioned that fungicides are used to control fungal infections rather than mentioning that fungicides are used to control the fungi which cause molds, rots, and other plant diseases. In part ii, some of the candidates were able to provide appropriate explanation about dyes using all the keywords such as 'a coloured compound capable of being fixed to a fabric', but were not able to recognise the colour of methyl orange under given condition. These responses mentioned that under acidic medium, methyl orange shows dark orange colour, rather than red or pink, and it appears to be colourless under basic conditions.

Example:



3) Herbicides
They are used at farms in agsiculture to save food from bectarias and insects.
4) Fungicides
They are used to avoid and cure funges infections.

DYES are compounds used to colour things, they are used in food, walls, door, cars etc.
Methyl orange is colourless in basic medium and has dark orange colour in acidic medium.

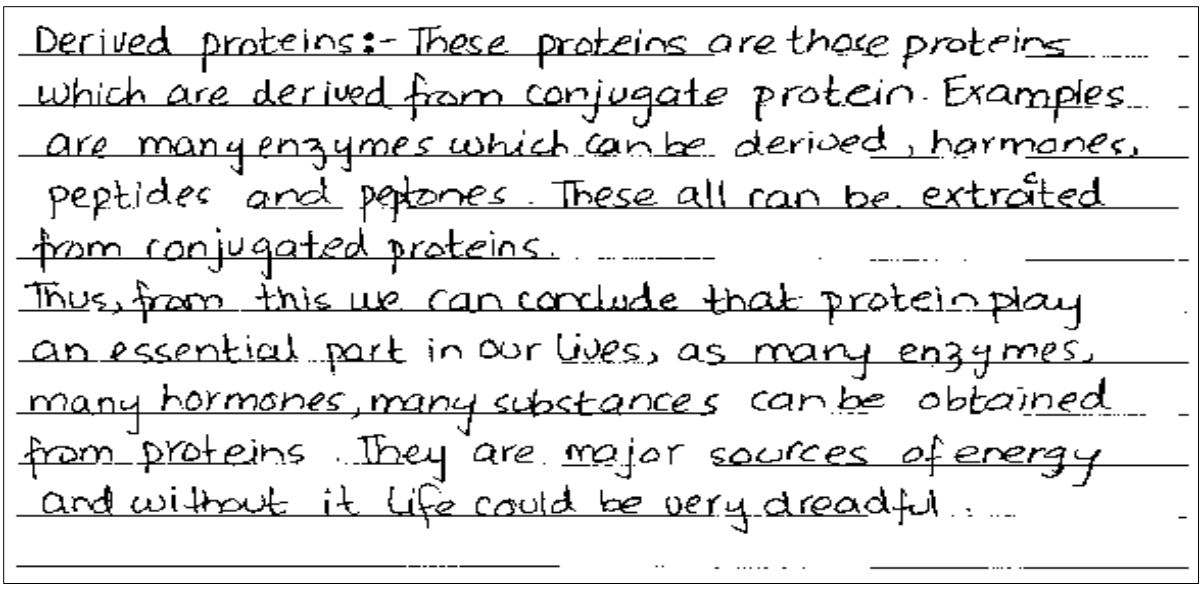
Question 11b:

i. What are proteins?
ii. Describe simple, conjugated and derived proteins. Give ONE example of each type.

Better responses mentioned few important points about protein such as amino acids are the major building block, and the presence of peptide linkage. In part ii, candidates explained simple protein in terms of their hydrolysis and highlighted albumin and globulin as their most common examples. For conjugate protein, candidates discussed about the non-protein group attached to amino acid chain and mentioned the examples of glycoprotein, phosphoprotein. They further described the derived proteins as proteins which are derived from simple and conjugated proteins and coated example of peptones.

Example:

Option b) i:- Proteins are actualy polymers made by fusion of many amino acids. These amino acids are linked by a peptide bond by condensation. Proteins are are usually main cource of energy for our life.
ii) Simple Proteins:- are those protein which can't be hydrolysed again and again. As they are made up of only two to three amino acids by peptide linkages eg:- Keratin, globulins etc.
Conjugated proteins:- When proteins combine with other prosthetic group (inorganic). This is conjugation example when phosphate combine with protains so it forms phospho proteins. When lipids and protein will combine, so it form lipoproteins.


Derived proteins:- These proteins are those proteins which are derived from conjugate protein. Examples are many enzymes which can be derived, harmones, peptides and peptones. These all can be extrated from conjugated proteins.
Thus, from this we can conclude that protein play an essential part in our lives, as many enzymes, many hormones, many substances can be obtained from proteins. They are major sources of energy and without it life could be very dreadful.

Weaker responses exhibited poor understanding of the concepts of biochemistry and defined protein and its types incorrectly and did not support their description with appropriate examples. Furthermore, candidates did not discuss that amino acids are linked through peptide bonds in protein. However, only few of these responses gave clear definition of protein. In part ii, candidates tried to explain simple protein with very general statements such as 'simple protein are quite simple molecule, basic building block of protein' etc. Some of these responses mixed the concept of protein and carbohydrate together and showed glucose as an example of simple protein. Similarly, in case of conjugate protein, candidates wrote that it contains two different monomer units or consist of chain of amino acid and represented fructose as an example. Finally, these responses struggled to describe the derived protein exactly, but indicated some basic knowledge about the derived protein. They identified derived proteins as a mixture of simple and conjugate protein, but mentioned starch as a wrong example.

Example:

option 'b'

i, proteins are complex molecules composed of Carbon, oxygen and amino acids. They help in providing energy to our body.

ii, Simple protein:-

Simple protein consists of basic building blocks of protein. They are simple molecules

e.g: glucose.

conjugated proteins:

They consist of chains of amino acids.

e.g: fructose.

Derived protein:

they are derived by combining conjugated and Simple proteins.

e.g = starch.